INCH-POUND

MIL-S-55590C(CR)
AMENDMENT 2
20 March 1991
SUPERSEDING
AMENDMENT 1
22 February 1973

MILITARY SPECIFICATION
SHELTER, ELECTRICAL EQUIPMENT
S-390( )/TRC-145( )

This amendment forms a part of MIL-S-55590C(EL), dated 19 January 1972, and is approved for use by the Communications-Electronics Command, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1 SPECIFICATIONS MILITARY:

Add: MIL-C-55427 Cable Assembly, Radio Frequency General Specification for

Delete: MIL-P-116 and MIL-S-55505.

PAGE 3

3.1 First Article of Production: Delete this paragraph and substitute the following:

"3.1 First Article. - When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3."

3.3.1 Delete "Power systems wiring and components" and substitute "Power systems and components installation".

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3.3.3 Change paragraph title to: "Signal (audio) system wiring and components installation." and add "(See 4.7.3)".

3.3.3.1 and 3.3.4.1 Change titles to: "Continuity of conductors."

3.3.4 Change paragraph title to: "Video and radio frequency (rf) systems wiring and components installation".

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3.3.5 Delete this paragraph and substitute the following:

"3.3.5 Cable Assemblies. - (See 4.4)

3.3.5.1 Cable Assemblies, Power and Audio Frequency. - All power and audio frequency cable and cord assemblies shall be fabricated in accordance with the drawings and meet the continuity, dielectric, insulation resistance, and pull-out requirements of MIL-C-3885.

3.3.5.2 Cable Assemblies, Radio Frequency. - Radio frequency cable assemblies required for operation of major items of equipment in 6.1 shall be fabricated in accordance with the drawings and meet the following requirements when tested in accordance with MIL-C-55427:

3.3.5.2.1 DC Resistance. - When tested as specified, the point to point resistance of the conductor shall not be greater than 1 ohm.

3.3.5.2.2 Pull-out Load for Terminations. - When tested as specified the cable assembly shall show no evidence of mechanical damage when a load of 25 pounds is applied.

3.3.5.2.3 Dielectric Withstanding Voltage. - When tested as specified the cable assembly shall show no evidence of breakdown with a test potential of 1500 volts RMS, 25-100 hertz.

3.3.5.2.4 Insulation Resistance. - When tested as specified the insulation resistance shall not be less than 100 megohms with a test potential of 500 volts DC.

3.3.5.2.5 Voltage Standing Wave Ratio. - When tested as specified the VSWR shall not exceed 1.20."

PAGE 6

3.11 Add: "(See 3.22)"

PAGE 7

3.13 Delete "(See 4.4)" and substitute "(See 3.22 and 4.4)".

3.14 Add "(See 3.22)"

3.15 Add "(See 3.22)"

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3.16 Add "(See 3.22)"

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3.17 Add "(See 3.22)"

3.18 Add "(See 3.22)"

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3.19 Add "(See 3.22)"

3.20 Add "(See 3.22)"

PAGE 10

3.22 Add "(See 4.17)" to the end of paragraph. Delete "3.21 Interchangeability" from list.

Add 3.23 as follows:

3.23 System Safety Engineering

3.23.1 Personnel Hazards. Personnel hazards shall be kept to a minimum. The criteria of MIL-STD-454, Requirement 1 and 8 shall be made applicable for tasks and efforts such as the selection of parts, the complete manufacturing and assembly process, and any production baseline configuration changes that may be implemented during the course of the contract. Verification that compliance with this requirement has been achieved shall be through a visual inspection (See 4.17).

3.23.2 Edge Rounding Exposed. Exposed edges shall be rounded to a minimum radius of 0.04 inch (1 mm), and exposed corners to a minimum of 0.5 inch (13 mm).

3.23.3 Radioactive Materials. Radioactive materials shall not be used (e.g. luminous dials/markings, electron tubes, surge arrestors and lenses).

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4.1 Responsibility for inspection: Delete second sentence and substitute the following:

"Except as otherwise specified in the contract or purchase order, the contractor may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government."

4.1.1 Contractor quality assurance: Delete this paragraph.

4.1.2 Government verification: Delete this paragraph.

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4.1.3 Accommodation and assistance: Delete this paragraph.

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4.2c, 4.2c(1) and 4.2c(2) Quality conformance inspection. Delete these paragraphs and substitute the following:

"4.2c Quality conformance inspection of equipment before packaging (See 4.5)."

4.3 First article inspection: Delete this paragraph and substitute the following:

"4.3 First article. - Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor. (See 3.1)

4.3.1 First article units. - The contractor shall furnish one (1) first article unit of the complete S-390( )/TRC-145(V).

4.3.2 First article inspection. - The first article inspection shall consist of inspections specified in Table I and shall be performed in the order specified therein.

4.3.3 First article data. The first article test plan and test report(s) shall be as required in the contract or purchase order.

4.3 Delete TABLE I with the notes at bottom and substitute the following:

TABLE I First Article Inspection

Inspections	Rqr Para	Test Para
1. Inspection covered by subsidiary documents	-	4.4
2. Interchangeability	3.21	4.16
3. Mounting Insert	3.10	4.14.1
4. Solderless-terminal-lug connectors	3.20.4	4.15
5. Visual and Mechanical	3.22	4.17
6. Power systems and components installation	3.3.1	4.7.1
7. Power distributional and operational function	3.3.2	4.7.2
8. Signal (audio) system wiring and components installation	3.3.3	4.7.3

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TABLE I First Article Inspection
(contd)

Inspections	Rqr Para	Test Para
9. Video and radio frequency (RF), system wiring and components installation	3.3.4	4.7.4
10. System operation	3.4	4.8
11. Vehicular transportation	3.5	4.9
12. Rail transportation	3.6	4.10
13. Flat drop	3.7	4.11.1
14. Rotational drop	3.7	4.11.2
15. Watertightness	3.8	4.12
16. Fording	3.9	4.13
17. Repeat 6 Power systems and components installation	3.3.1	4.7.1
18. Repeat 7 Power distributional and operational function	3.3.2	4.7.2
19. Repeat 8 Signal (audio) system wiring and components installation	3.3.3	4.7.3
20. Repeat 9 Video and radio frequency (RF), system wiring and components installation	3.3.4	4.7.4
21. Repeat 10 System operation	3.4	4.8

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4.5 Change title to: "Quality conformance inspection of equipment before packaging." and delete "(See 4.1.2)" and the last sentence.

4.5.1 Group A inspection: Delete this paragraph and substitute the following:

"4.5.1 Group A inspection. - Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table II. Lots shall be formed only from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and AQL's indicated in Table II.

4.5.1.1 Order of inspection within Group A. - Group A inspection shall be performed in the order listed in Table II."

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Delete TABLE II Lot Formation

Replace TABLE III with the following:

TABLE II - Group A Inspection

Inspection	Rqr Para	Insp Para	AQL % Def Major	Minor
Visual and Mechanical	3.22	4.17	1.0	4.0
Electrical installation				**
Power systems and component installation				
Continuity of conductors	3.3.1.1	4.7.1.1		
Dielectric breakdown voltage	3.3.1.2	4.7.1.2		
Insulation resistance	3.3.1.3	4.7.1.3		
Power distributional and operational function	3.3.2	4.7.2 thru 4.7.3.2		
Signal (audio) system wiring and components installation				
Continuity of conductors	3.3.3.1	4.7.3.1		1.0 for the entire group of power, signal and video wiring and waveguide assemblies
Dielectric breakdown voltage	3.3.3.2	4.7.3.2		
Insulation resistance	3.3.3.3	4.7.3.3		
Video and radio frequency (RF), system wiring and components installation				
Continuity of conductors	3.3.4.1	4.7.4.1		
Dielectric breakdown voltage	3.3.4.2	4.7.4.2		
Insulation resistance	3.3.4.3	4.7.4.3		
VSWR	3.3.4.4	4.7.4.4		

** All defects are major

4.5.2 Group B inspection. - Delete this paragraph and substitute the following:

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"4.5.2 Group B inspection. - Group B inspection shall normally be performed on inspection lots that have passed Group A inspection, except for Interchangeability and Mounting Insert Inspections which are performed prior to Group A. This inspection shall conform to Table III and to the special inspection levels of table I of MIL-STD-105."

4.5.2.1 Group B sampling plans. - Replace "TABLE IV" with "TABLE III."

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4.5.2.2 Order of inspection within Group B. - Delete "in which they appear" and substitute "listed in TABLE III."

Change table number from "TABLE IV" to "TABLE III" and add the following note to TABLE III Group B inspection:

"NOTE: The order of inspection of Group B with respect to Groups A and C shall conform to the order specified in 4.3. The order of inspection within Group B shall be as in 4.5.2.2."

4.5.3 Group C Inspection. - In this paragraph in title of Group C Inspection Table and in note at bottom of table, change "TABLE V" to "TABLE IV".

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4.5.3.2 Change title of paragraph from "Sampling for inspection of construction evaluation" to " Sampling for inspection of construction verification."

4.5.3.5 Delete this paragraph and replace with the following:

"4.5.3.5 Group C failures. - Actions required relative to Group C failures shall be as specified in the contract or purchase order."

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4.5.4 In the first line delete "Group B and", in the third line delete "Group B" and "or both," and in the fourth line change "Group C" to "Group A".

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4.7.1.1 Continuity of conductors. - In first sentence delete "which meets" and substitute "tests to meet".

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4.7.1.2 Dielectric breakdown voltage. - In first sentence delete "which meets" and substitute "tests to meet".

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4.7.1.3 Insulation resistance. - In first sentence delete "which meets" and substitute "tests to meet".

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4.7.3.1 Continuity of conductors. - In first sentence delete "which meets" and substitute "tests to meet".

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4.7.3.2 Dielectric breakdown voltage. - In first sentence delete "which meets" and substitute "to meet".

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4.7.3.3 Insulation resistance. - In first sentence delete "which meets" and substitute "tests to meet".

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4.7.4 Change to the following:

"4.7.4 Video and radio frequency (RF) system wiring and components installation."

4.7.4.1 Continuity of conductors. - In first sentence delete "which meets" and substitute "tests to meet".

Change "TABLE VI" to "TABLE V".

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4.7.4.2 Dielectric breakdown voltage. - In first sentence delete "which meets" and substitute "to meet".

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4.7.4.3.a Change "4.7.4.2a" to "4.7.4.2.c"

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4.10 Change title to "Rail transportation test", in the sixth line add "(contractor furnished)" after slings, and add after the sentence ending in line fifteen, the following new sentence - "All cars shall have standard (rigid) draft gear".

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4.10.1 Test flatcar. - In line seven delete "power" and in line ten delete "not".

4.10.2.a Delete "springly" and substitute "springy".

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4.14 Mounting insert tests. - Replace "Preproduction" with "First article" in 4.14.1 and table.

Change "TABLE VII" to "TABLE VI".

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Change "TABLE VIII" to "TABLE VII".

In table add the following to major defects:

"13. Center pin of rf cable assemblies improperly crimped or soldered, loose or protruding".

4.18 Quality conformance inspection of preparation for delivery. Delete.

4.19 Rough handling test (preparation for delivery). Delete.

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5. and 5.1 Delete and substitute the following:

"5. PACKAGING

5.1 Packaging requirements. The packaging requirements for the desired level(s) of protection shall be as specified by the acquisition activity."

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Preparing Activity:
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